When we want to say a number obtained by adding 5 and 5, it is easier to express it by using brackets like this: (5 + 5).

(17 - 7) means a number obtained by subtracting 7 from 17.

(5 x 2) means a number obtained by multiplying 5 by 2.

(20 ÷ 2) means a number obtained by dividing 20 by 2.

Arrangement of numbers like (5 + 5), (17 - 7), (5 x 2) and (20 ÷ 2) are called expressions. The value of each of these expressions is 10, which means all these expressions are equal to each other.

Hence, we can also write this as,

(5 + 5) = (17 - 7) = (5 x 2) = (20 ÷ 2)

An expression such as (5 + 5) = (17 - 7) or (5 + 5) = (20 ÷ 2) is called an 'equality'.

5 + 2 = 7, 8 x 1 = 8 are also equalities.

Exercise 4.1

1. Using brackets, write three pairs of numbers whose sum is 12. Use them to write three equalities.
2. Find four pairs of numbers, one for each of addition, subtraction, multiplication and division that make the number 16. Write the equalities for each of them.

Inequality

The value of the expressions (6 + 3) and (6 x 3) are 9 and 18 respectively. It means the above two expressions are not equal.

We can consider another example. Take two numbers 4 and 5. We know that 4 is not equal to 5. But, still we can relate those two numbers by a relationship.

If two expressions or numbers are not equal, one of them is greater or smaller than the other. To show 'greater than' and 'lesser than', we use the symbols '>' and '<' respectively.

This kind of representation is called an 'inequality'.

Let us consider another example, the value of (9 - 5) is 4 and the value of (25 ÷ 5) is 5 and we know that 4 < 5. So, the relation between the expressions (9 - 5) and (25 ÷ 5) can be shown as (9 - 5) < (25 ÷ 5).

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